

1.

Item	Factor of Production	Explanation
A. A teacher working in a school	Labour	Human effort used in production
B. A sewing machine in a textile factory	Capital	Man-made tool used in production
C. A plot of land used for farming	Land	Natural resource
D. An investor starting a new tech startup	Entrepreneurship	Organizer and risk-taker in production
E. A river used for hydroelectric power generation	Land	Natural resource

2.

Decision	Opportunity Cost	Explanation
A. Watching a movie instead of studying	The study time or potential grade improvement	Gave up study time
B. Baking cakes instead of bread	The bread that could have been baked	Alternative forgone
C. Investing in stocks instead of fixed deposit	Interest income from fixed deposit	Missed return
D. Buying a smartphone instead of laptop	The laptop	Foregone choice
E. Taking unpaid leave to go on vacation	Lost income	Wages forgone

4.

Country Wheat Cloth

A	10	20
B	5	10

(A) *Absolute Advantage:*

- **Country A** has an absolute advantage in both goods (produces more of both with same resources).

(B) *Opportunity Cost:*

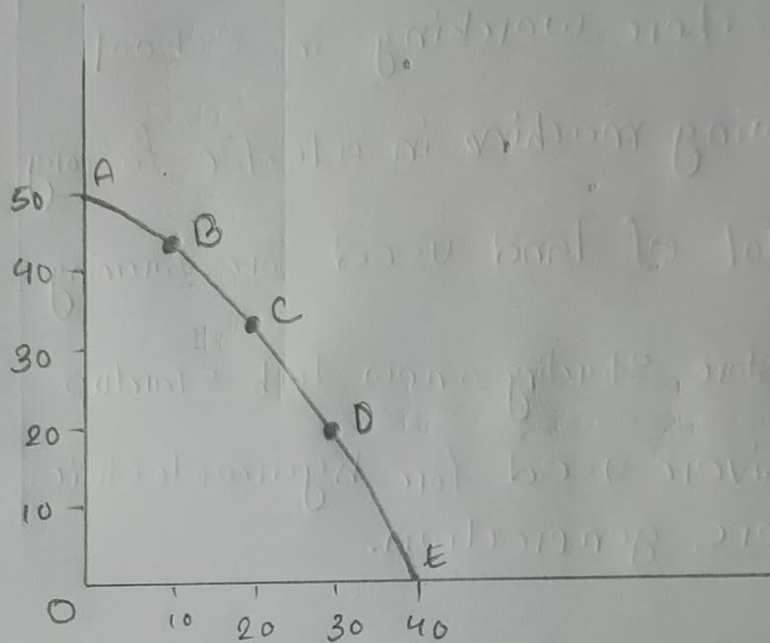
Country	1 Wheat costs	1 Cloth costs
A	2 Cloth / 10	$\frac{1}{2}$ Wheat
B	10/5	$\frac{1}{2}$ Wheat

So they are identical — **no comparative advantage** (opportunity cost same).

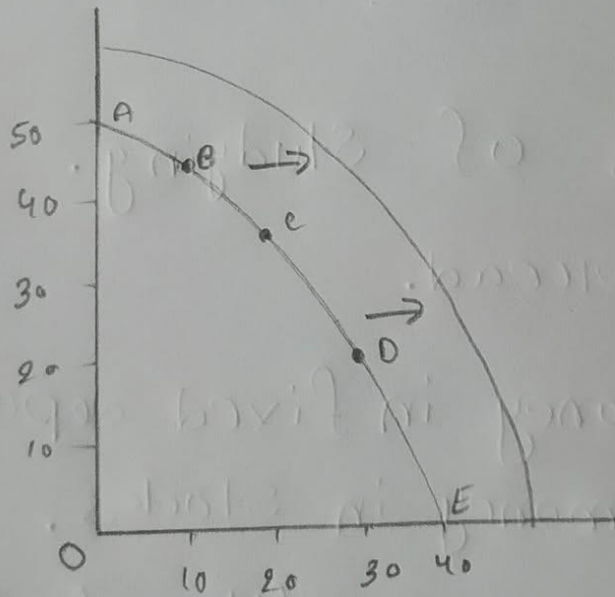
3.

3

A.



B.



There will be an outward shift of economic growth due to improvement in technology.

5.

Price Qd Qs

10 50 10

20 40 20

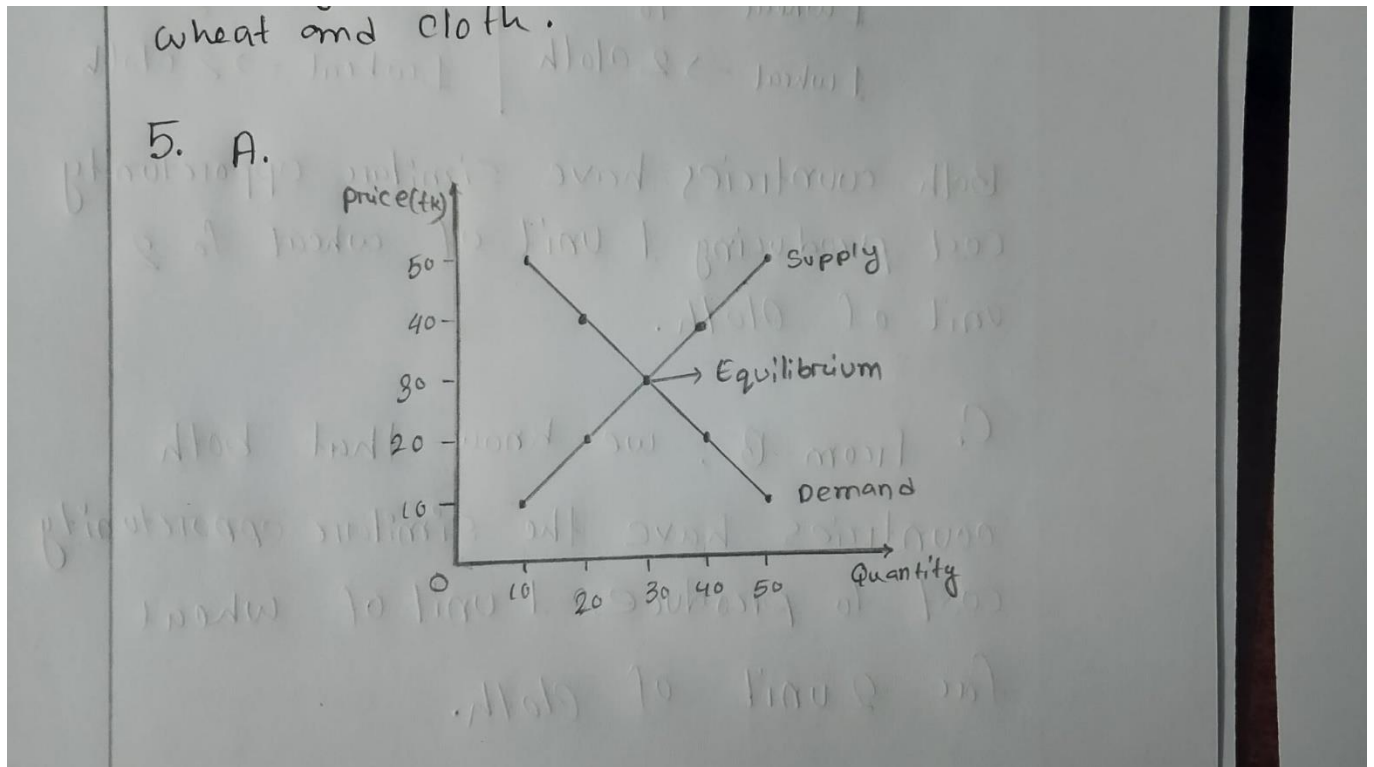
30 30 30

40 20 40

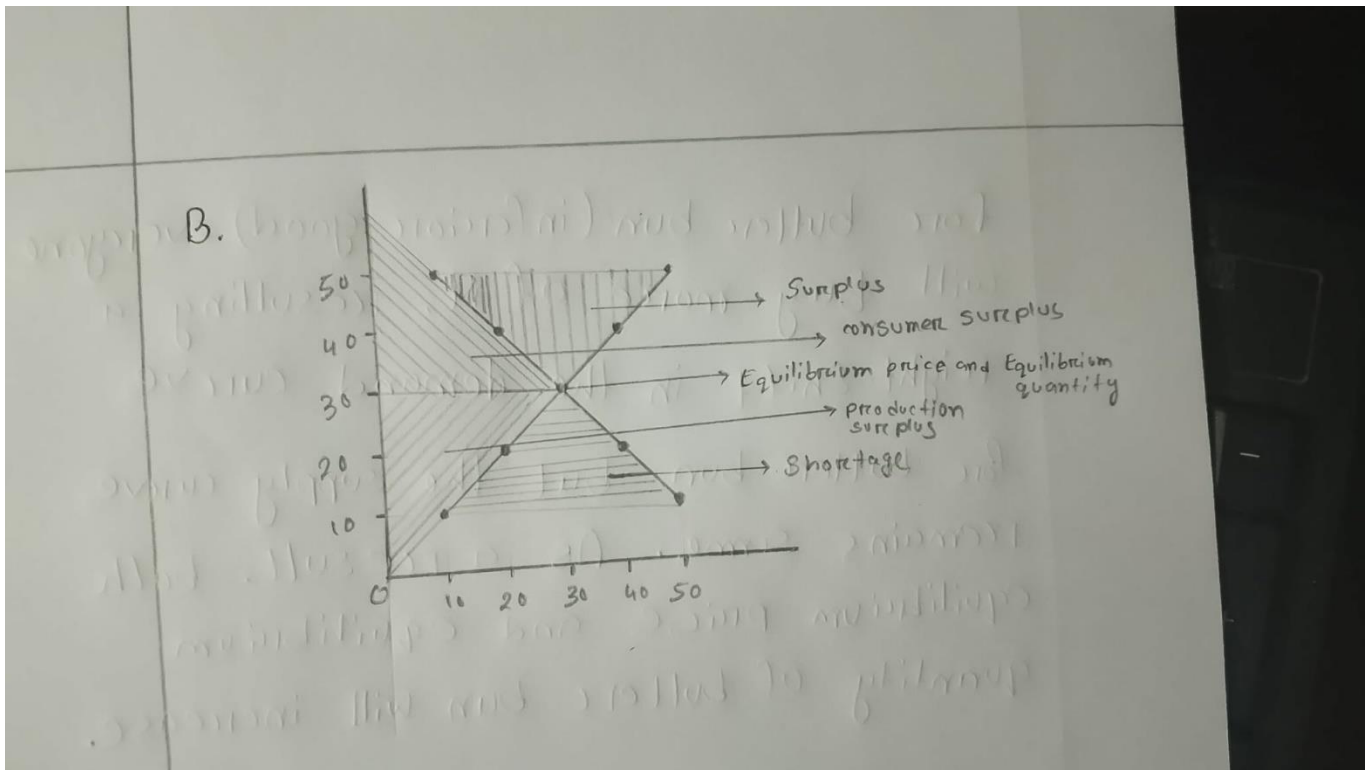
50 10 50

(A) Graph:

- Downward sloping demand, upward sloping supply.
- Equilibrium at $P = 30$, $Q = 30$.



(B) Labels:



- **Equilibrium Price:** 30
- **Equilibrium Quantity:** 30
- **Surplus:** Above equilibrium ($Q_s > Q_d$)
- **Shortage:** Below equilibrium ($Q_d > Q_s$)
- **Consumer Surplus:** Area above P_e and below demand curve
- **Producer Surplus:** Area below P_e and above supply curve

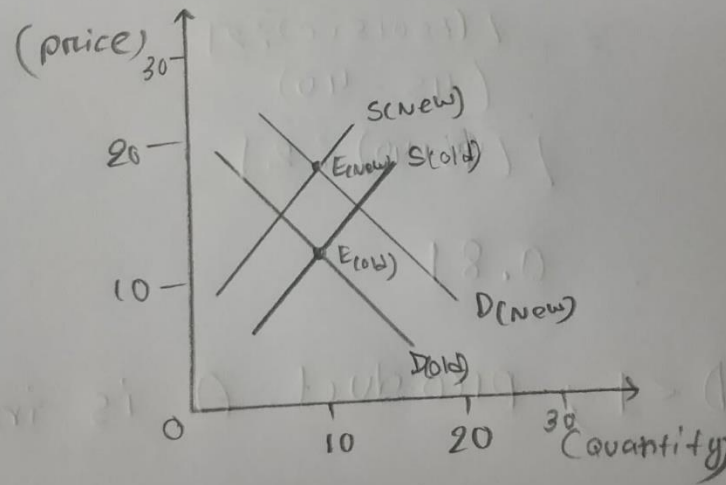
6.

dogs (normal) & Butter Buns (inferior)

- Income $\downarrow$ (due to tax $\uparrow$)
- Demand for **normal goods (hotdogs)** $\downarrow \rightarrow$ **Price and quantity fall.**
- Demand for **inferior goods (butter buns)** $\uparrow \rightarrow$ **Price and quantity rise.**

7.

profit. So, the price of bicycles will decrease and there will be a left-shift in the supply curve.



Situation:

- Fuel prices $\uparrow \rightarrow$ demand for electric bicycles $\uparrow$ (rightward shift in demand)
- Raw material cost $\uparrow \rightarrow$ supply of electric bicycles $\downarrow$ (leftward shift in supply)

Diagram:

- Demand curve shifts **rightward**, supply curve shifts **leftward**.

Effect:

- Price: **Increases**
- Quantity: **Uncertain** (depends on magnitude of shifts)

8.

Product June (P,Q) Sept (P,Q)

A	40, 500	45, 550
B	55, 800	60, 780

(A) Price Elasticity of Demand (PED)

$$PED = \frac{\% \Delta Q}{\% \Delta P} = \frac{(550 - 500) / ((550 + 500) / 2)}{(45 - 40) / ((45 + 40) / 2)} = \frac{50 / 525}{5 / 42.5} = \frac{0.0952}{0.1176} = 0.81$$

PED = 0.81 (Inelastic)

→ Demand not very responsive to price.

(B) *Cross Price Elasticity of Demand (XED)*

Price of B ↑ from 55 → 60

Quantity of A ↑ from 500 → 550

$$XED = \frac{\% \Delta Q_A}{\% \Delta P_B} = \frac{(550 - 500) / 525}{(60 - 55) / 57.5} = \frac{0.0952}{0.0870} = 1.09$$

XED = +1.09 (Positive, substitutes) →

Product A and B are **substitute goods**.